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Final Narrative (Decision Memo)

Retail Inventory Optimization: Forecasting and Replenishment Strategy

1. Business Goal and KPIs

Retail businesses face a continuous challenge of balancing product availability with inventory holding costs. Stockouts result in lost sales and reduced customer satisfaction, while excessive inventory leads to higher storage and capital costs. The objective of this project is to **reduce stockouts while maintaining efficient inventory levels** across multiple stores and product SKUs.

The **North Star metric for this project is Service Level (Fill Rate)**, which measures the proportion of customer demand that is successfully fulfilled from available inventory.

Fill Rate=Units SoldTrue DemandFill\ Rate = \frac{Units\ Sold}{True\ Demand}Fill Rate=True DemandUnits Sold

Improving fill rate directly reflects better product availability and reduced lost sales.

To support this metric, several operational KPIs were used to evaluate system performance:

- Forecast Accuracy (measured using WAPE)
- Stockout Rate
- Fill Rate
- Inventory Turns (proxy)
- Days of Inventory on Hand (DOH)
- Overstock Rate
- Holding Cost Proxy
- Lost Sales Proxy

These metrics collectively provide insight into both **customer service performance and inventory efficiency**.

2. Data and Methodology

The analysis was conducted using historical retail data containing store-level sales, inventory records, product information, and calendar indicators.

The workflow consisted of three major stages:

Data Processing (ETL)

The data was cleaned and standardized through an ETL pipeline. Key steps included:

- Standardizing product categories and removing invalid records
- Cleaning inventory data and handling missing values
- Creating daily **store-SKU level fact tables** for sales and inventory
- Calculating **true demand**, accounting for stockouts
- Integrating product attributes such as shelf life and minimum order quantities

This process produced structured datasets suitable for forecasting and replenishment analysis.

Demand Forecasting

The project evaluated baseline forecasting approaches to predict demand for the next **28 days**.

The primary model selected was a **7-day moving average model**, which captures short-term demand trends and weekly seasonality patterns common in retail data.

The model works by forecasting daily demand as the average demand observed over the previous seven days.

Forecast accuracy was evaluated using **Weighted Absolute Percentage Error (WAPE)**, which is preferred for retail demand data because it remains stable even when demand values are small.

The moving average model provided a simple and interpretable baseline forecast suitable for operational planning.

Replenishment Policy

Using the demand forecasts, a replenishment policy was implemented for each **store–SKU pair**.

The policy calculates the following components:

Safety Stock

Safety stock accounts for demand variability and ensures product availability during supply lead times.

Safety Stock=Z×σdemand×Lead TimeSafety\ Stock = Z \times \sigma_{demand} \times \sqrt{Lead\ Time}Safety Stock=Z×σdemand×Lead Time

where the service level target corresponds to a Z-score.

Reorder Point (ROP)

ROP=*Avg Daily Demand*×*Lead Time*+*Safety Stock*ROP=Avg Daily Demand×Lead Time+Safety Stock

This represents the inventory level at which a new order should be triggered.

Recommended Order Quantity

The recommended order quantity is calculated using a **target days-of-cover policy**, ensuring sufficient stock until the next replenishment cycle.

Operational constraints were incorporated into the policy:

- Minimum Order Quantity (MOQ)
- Shelf-life limitations for perishable products
- Store storage capacity limits

These constraints ensure that replenishment decisions remain **practical and operationally feasible**.

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3. Key Insights

Several important insights emerged from the analysis:

1. Significant stockout risk exists across multiple store–SKU combinations.
Several products exhibited very low days of inventory cover, indicating high risk of stockouts if replenishment decisions are not optimized.

2. Demand patterns exhibit clear weekly seasonality.
Demand tends to follow recurring weekly patterns, which is effectively captured by the moving average model.

3. Stockouts contribute significantly to lost sales.
When inventory levels reach zero, actual sales are constrained by supply rather than customer demand.

4. Some stores contribute disproportionately to stockout occurrences.
This suggests potential operational differences in demand patterns or replenishment effectiveness across stores.

5. Overstock risk is relatively limited in the current dataset.
Most issues are associated with insufficient inventory rather than excessive inventory levels.

6. Demand variability differs across product categories.
Certain categories show higher volatility, requiring larger safety stock buffers.

7. Replenishment policies can significantly reduce stockout risk while maintaining controlled inventory levels.

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4. Forecast Performance and Model Choice

Multiple baseline forecasting approaches were explored, including moving average and other simple models. The **7-day moving average model** was selected as the primary forecasting method due to its simplicity, interpretability, and ability to capture weekly demand patterns.

Forecast performance was evaluated using WAPE, which measures the total absolute forecast error relative to total demand.

While the baseline model provides reasonable demand estimates, it does not explicitly incorporate promotional or holiday effects. These factors were instead visualized in the dashboard to help interpret demand spikes.

Future improvements could include **regression-based models incorporating promotional and calendar variables**.

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5. Replenishment Plan

The proposed replenishment strategy uses a **continuous review inventory policy**.

For each store–SKU combination:

1. Demand forecasts estimate expected demand over the lead time.
2. Safety stock accounts for demand uncertainty.
3. A reorder point is calculated.
4. When inventory falls below the reorder point, a replenishment order is triggered.

The recommended order quantity is constrained by operational factors such as:

- minimum order quantities
- shelf-life limits
- store storage capacity

This approach ensures that inventory levels are maintained at sufficient levels to prevent stockouts while avoiding excessive inventory accumulation.

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6. 30-Day Impact Estimation

To estimate the business impact of the proposed replenishment strategy, three scenarios were analyzed over a **30-day horizon**.

Base Scenario

Assumes a **20% reduction in stockout days** due to improved replenishment decisions.

Estimated impact:

- **492,149 units of lost sales avoided**
- **₹130,720,516 revenue recovered**
- **₹24,493,836 increase in inventory value**

Best Case Scenario

Assumes **35% stockout reduction**.

Estimated impact:

- **861,261 units of lost sales avoided**
- **₹228,760,902 revenue recovered**
- **₹36,740,753 increase in inventory value**

Worst Case Scenario

Assumes **10% stockout reduction**.

Estimated impact:

- **246,075 units of lost sales avoided**
- **₹65,360,258 revenue recovered**
- **₹15,308,647 increase in inventory value**

These projections indicate that even modest improvements in replenishment decisions can generate significant revenue recovery.

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7. Risks, Limitations, and Next Steps

While the proposed solution provides meaningful operational improvements, several limitations exist.

First, the forecasting model used is a baseline model and does not explicitly incorporate promotional or holiday effects, which may influence demand spikes.

Second, the analysis assumes consistent lead times and does not model potential supply disruptions.

Third, the impact estimates rely on simplified assumptions regarding stockout reduction and demand recovery.

Future improvements could include:

- implementing advanced forecasting models such as regression or machine learning models
- incorporating promotional and holiday calendars directly into forecasting models
- optimizing inventory policies using simulation-based approaches
- integrating the replenishment system with real-time inventory monitoring systems.

Conclusion

This project demonstrates how demand forecasting combined with structured replenishment policies can significantly improve retail inventory management. By focusing on service level improvement and data-driven replenishment decisions, the proposed approach can reduce stockouts, recover lost sales, and maintain efficient inventory levels across stores.

The results highlight the potential value of implementing analytical decision support systems in retail supply chain operations.